

P 1 Conservation and dissipation of energy

1.1 Changes in energy stores

Learning objectives

After this topic, you should know:

- the ways in which energy can be stored
- how energy can be transferred
- the changes in energy stores that happen when an object falls
- the energy transfers that happen when a falling object hits the ground without bouncing back.

On the move

Cars, buses, planes, and ships all use fuels as chemical energy stores. They carry their own fuel. Electric trains use energy transferred from fuel in power stations. Electricity transfers energy from the power station to the train.



Figure 1 The French Train à Grande Vitesse electric train can reach speeds of more than 500 km/hour

Energy can be stored in different ways and is transferred by heating, waves, an electric current, or when a force moves an object. Here are some examples:

- Chemical energy stores include fuels, foods, or the chemicals found in batteries. The energy is transferred during chemical reactions.
- Kinetic energy stores describe the energy an object has because it is moving.
- Gravitational potential energy stores are used to describe the energy stored in an object because of its position, such as an object above the ground.
- Elastic potential energy stores describe the energy stored in a springy object when you stretch or squash it.
- Thermal energy stores describe the energy a substance has because of its temperature.

Energy can be transferred from one store to another. In a torch, the torch's battery pushes a current through the bulb. This makes the torch bulb emit light, and also get hot (Figure 2).

When an electric kettle is used to boil water, the current in the kettle's heating element transfers energy to the thermal energy store of the water and the kettle.

When an object is thrown into the air, the object slows down as it goes up. Here, energy is transferred from the object's kinetic energy store to its gravitational potential energy store.

You can show the energy transfers by using a flow diagram:

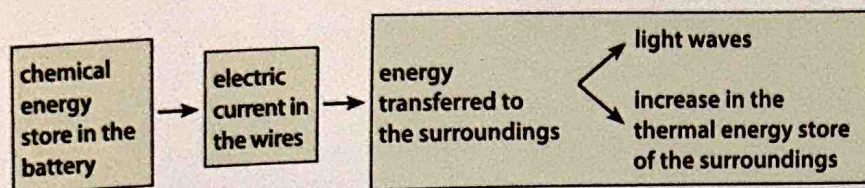


Figure 2 Changes in energy stores in a torch lamp

Energy transfers

When an object starts to fall freely, it speeds up as it falls. The force of gravity acting on the object causes energy to be transferred from its gravitational potential energy store to its kinetic energy store.

Look at Figure 3. It shows an object that hits the floor with a thud. All of the energy in its kinetic energy store is transferred by heating to the thermal energy store of the object and the floor, and by sound waves moving away from the point of impact. The amount of energy transferred by sound waves is much smaller than the amount of energy transferred by heating.

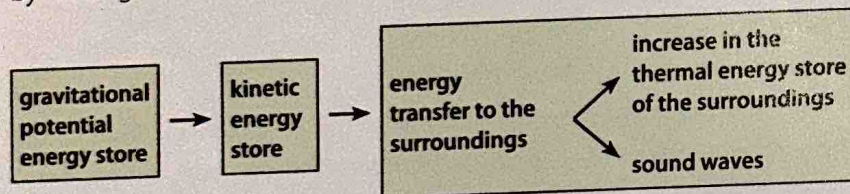


Figure 4 An energy transfer diagram for an object when it falls and when it hits the ground

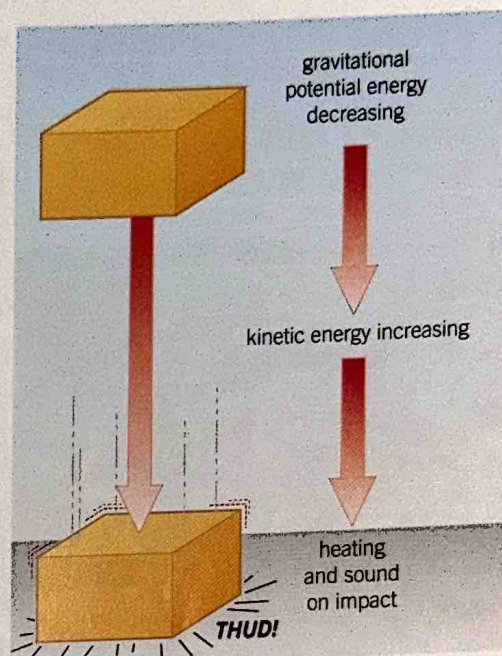


Figure 3 An energetic drop. On impact, energy is transferred to the thermal energy store of the surroundings by heating and by sound waves

Key points

- Energy can be stored in a variety of different energy stores.
- Energy is transferred by heating, by waves, by an electric current, or by a force when it moves an object.
- When an object falls and gains speed, its store of gravitational potential energy decreases and its kinetic energy store increases.
- When a falling object hits the ground without bouncing back, its kinetic energy store decreases. Some or all of its energy is transferred to the surroundings – the thermal energy store of the surroundings increases, and energy is also transferred by sound waves.

- Describe the changes to energy stores that take place when:
 - a ball falls in air [2 marks]
 - an electric heater is switched on. [2 marks]
- List two different objects you could use to light a room if you have a power cut. For each object, describe the energy transfers and changes to energy stores that occur when it lights up the room. [4 marks]
 - Which of the two objects in **a** is:
 - easier to obtain energy from? [1 mark]
 - easier to use? [1 mark]
- Describe the changes in energy stores of an electric train as it:
 - moves up a hill at constant speed [2 marks]
 - approaches a station and brakes to a halt. [2 marks]
- Describe the changes in energy stores that take place when food is heated in a microwave oven. [2 marks]